

100 Days of Azure DevOps

Day 71 Handout — Metrics are the pulse, logs are the sentences, activity is who changed ARM — diagnostics are opt-in

8 - Monitoring & Observability · Personal lab only · Educational content · Not a sales pitch

Architecture A — four signal families (Monitor is the flashlight)

Signal	What it answers	Where it lives
Metrics	CPU, availability, queue length — numbers on a time axis	Platform metrics on the resource. Near-real-time, cheap-ish
Logs	The story in rows you query with KQL	Log Analytics workspace, only after diagnostic settings (or agents) send them
Activity log	Control plane: who PUT/DELETE this RG / vault / AKS	Subscription activity. Not the app's 500s
Diagnostic settings	The pipe: resource → workspace / storage / event hub	Opt-in. Without them the workspace is an empty room
Insights blades	App Insights, Container Insights, VM Insights — packaged views	Still Monitor. They do not patch the hole

Architecture B — turn on a light without buying a SIEM

Constraint	Default today	Skip / delay
One resource	Key Vault AuditEvent or App Service HTTP logs → LAW	Every resource in the sub 'for completeness'
Retention	Short (workspace default / 30 days class) for a lab	730-day retention as a personality trait — that is a storage bill
Cost already tight	Map the four signals on paper; skip the workspace	Leaving diagnostics on a chatty resource all month
'It is down'	Pick one signal you would actually look at	A dashboard with no diagnostic settings behind it
Alerts	Tomorrow+ (Day 74). Today is the pipe	Alert rules with no destination yet

Architecture C — pitfalls that start a war room with screenshots and feelings

Pitfall	What you see	Fix
No diagnostic setting	Workspace queries return zero; people argue	Enable logs to LAW on one lab resource
Activity log as app log	You know who deleted the webapp, not why it 500'd	Need App Insights / resource logs for that
Metrics without dimensions	Average CPU looks fine, one instance is on fire	Split by instance when the resource has more than one
Infinite retention	Cost Analysis surprise	Set a finite cap; lab can be minimal
Monitor as the fix	Pretty chart of 500s, still 500s	Flashlight ends the argument about reality. Patch is later

One-liner to remember

If you cannot see it, you cannot fix it — Monitor is the flashlight, not the fix. Turn on diagnostics before you argue about which wall is wet.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal subscription. One lab resource (Key Vault from Phase 7 or an App Service). Short retention. No employer workspace.

- 1 Portal → Monitor → Overview. Click through Metrics, Activity log, Logs. Write one sentence each: pulse vs diary vs sentences.
- 2 Create or reuse a Log Analytics workspace in a lab RG (law-lab) if cost allows. If you skip, still complete the paper map and say why.
- 3 On one resource (Key Vault or App Service) → Diagnostic settings → send Audit / HTTP logs + AllMetrics to law-lab. Category vs category group: pick what the blade offers.
- 4 Generate a control-plane event (tag the resource). Activity log should show your user. That is not an app failure.
- 5 Write docs/monitor-day71.md: which signal you would use to detect 'it is down', retention choice, resource id of the diagnostic setting.
- 6 Do not onboard the whole subscription. Do not start a 90-day retention experiment. KQL is tomorrow.

Starter CLI — diagnostic setting to a workspace

```
az monitor log-analytics workspace create -g rg-day71 -n law-lab -l eastus
LAW=$(az monitor log-analytics workspace show -g rg-day71 -n law-lab --query id -o tsv)
KV=$(az keyvault show -n <uniquekv> --query id -o tsv)
az monitor diagnostic-settings create --name kv-to-law --resource $KV --workspace $LAW \
  --logs '[{"category":"AuditEvent","enabled":true}]' \
  --metrics '[{"category":"AllMetrics","enabled":true}]'
# Activity log is subscription-level (Monitor → Activity). It is not this pipe.
```

Done checklist

- Can separate metrics, logs, and activity log without mixing them
- One diagnostic setting exists — or a written cost skip plus the paper map
- Knows diagnostics are opt-in
- docs/monitor-day71.md names the 'it is down' signal

Tomorrow — Day 72

Log Analytics and KQL — ask a precise question, stop screenshotting 10,000 rows